to get the red and black wires for positive and negative supplies of the circuit. Cut the green and white wires of the USB cable and solder the red and black wires to the PCB. Use the USB plug at the other end to draw power from the USB socket. Fix the unit near the keyboard so you can see the key buttons easily. ●

SOLID-STATE TWO-WAY SWITCH



■ PRATIK PANCHAL

Two-way switches, more profoundly known as staircase switches, are widely used when a load has to be controlled from two ends of the location. For example, a staircase lamp that needs to be controlled from both ends of the staircase, or a hallway or corridor light that needs to be controlled from both ends of the halfway.

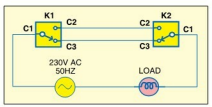


Fig. 1: Schematic for two-way witch wiring

Fig. 1 shows the two-way switching arrangement. K1 and K2 are single-pole, double-throw (SPDT) switches that are installed at the two ends of a location. Fig. 2 shows the switching mechanism for 'on' and 'off' states of the load.

In home appliance automation, switching is usually controlled by a microcontroller. The output of the microcontroller is used to control the operation of the relay for switching the electrical appliance.

Fig. 3 shows the complete circuit for two-way switching. The appliance can either be controlled by mechanical switch K2 or solid-state switch K1. Switch K1 can be either electronically controlled or computer-controlled.

The 230V AC live (L) wire is connected to C1 terminal of switch K2 through bulb B1. When a high logic (5V) is applied at pin 1 of IC2 through the control signal from the microcontroller, the internal LED of opto-isolator IC2 (MOC3020) glows. As a result, TRIAC1 fires and the phase voltage (230V AC) becomes available at terminal C2 of K1, causing bulb B1 to glow. LED1 indicates that opto-isolator IC2 is active. Hex inverter N1 (IC1) ensures that either IC2 or IC3 is activated at any given time.

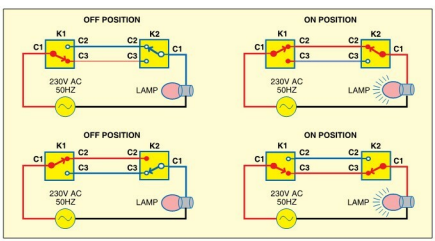


Fig. 2: Working of two-way switch circuitry

When a low logic (0V) is applied to pin 1 of IC2 through the control signal from the microcontroller, opto-isolator IC2 deactivates but opto-isolator IC3 activates through the inverter output (pin 12) of gate N1 (IC 7404). This is indicated by LED2. Thus when the control signal from the microcontroller is high/low, the phase voltage (230V AC) is present on C2/C3 terminal of K1. This

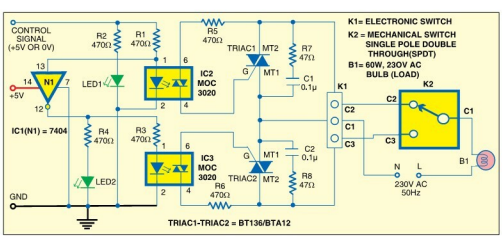


Fig. 3: Interfacing of solid-state two-way switch (K1) with mechanical two-way switch (K2)